

Professor-in-a-Box

A Personalized AI Tutor Powered by QLoRA and a Cloud Data Pipeline

Final ISA Presentation · DATA 298A

Fall 2025 – Final ISA Demo

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End-to-End Cloud Pipeline · QLoRA Fine-Tuning · Real-Time Tutor

Our goal was to build an AI tutor that understands student reasoning and explains concepts the way a human instructor does.

Project Vision: Why We Built This

Computer Science education often lacks personalized guidance for deep conceptual understanding. Complex topics like recursion and debugging require tailored explanations that static platforms cannot offer. We envisioned an AI tutor, 'Professor-in-a-Box', to provide real-time, personalized instruction, clarify concepts, and guide students through debugging with human-like reasoning.

Challenges in CS Learning

- Over-emphasis on problem-solving over conceptual depth
- Difficulty in getting personalized explanations
- Lack of support for guided debugging

AI Tutor Capabilities

- Real-time conceptual clarification
- Guided debugging pathways
- Reasoning-first explanations
- Scalable personalized tutoring

This semester, we completed the full ISA pipeline lifecycle: data ingestion → transformation → warehouse → monitoring → model evaluation → demo.

Semester Accomplishments (What We Did End-to-End)

01

Automated Cloud Pipeline (Airflow DAGs)

Fully automated Airflow pipeline for data ingestion and transformation.

03

Curated Instruction Dataset

Curated a 5,000-sample CS instruction dataset.

05

Real-Time Inference (<200ms)

Achieved real-time inference (<200ms) for interactive tutoring.

02

Versioned Storage Layer (S3 Warehouse)

Designed a versioned S3 data warehouse with industry-standard practices.

04

QLoRA Fine-Tuning (Qwen2.5-Coder-7B)

Fine-tuned Qwen2.5-Coder-7B with QLoRA on consumer hardware.

06

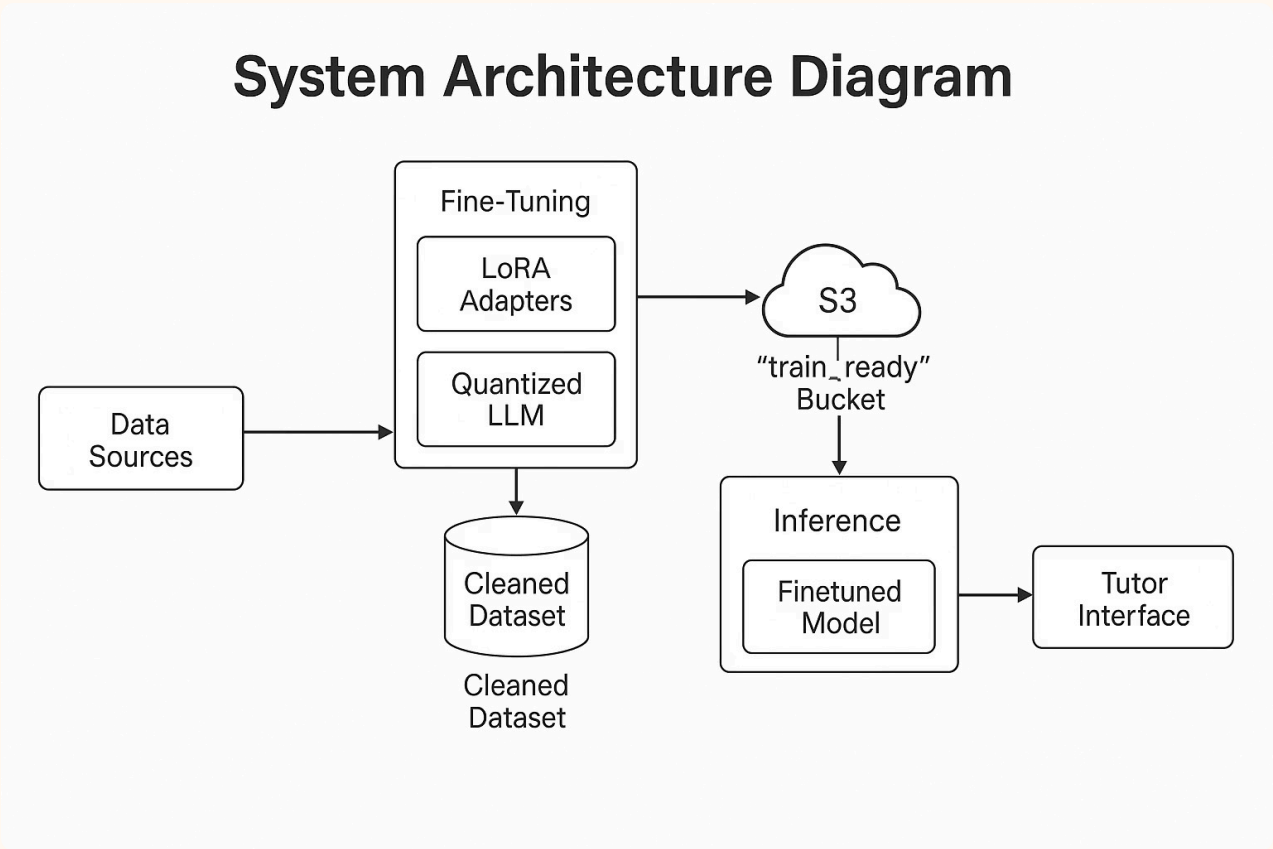
Complete ISA Demo (End-to-End)

Delivered a full ISA demo, covering the entire pipeline.

This architecture forms the foundation of our end-to-end ISA pipeline, connecting data ingestion through model serving.

End-to-End System Architecture

Our architecture integrates:



This design allows the system to be modular, scalable, and reproducible across training cycles.

Ingestion is fully automated, recoverable, and validated at every step.

Data Ingestion Pipeline (Airflow)

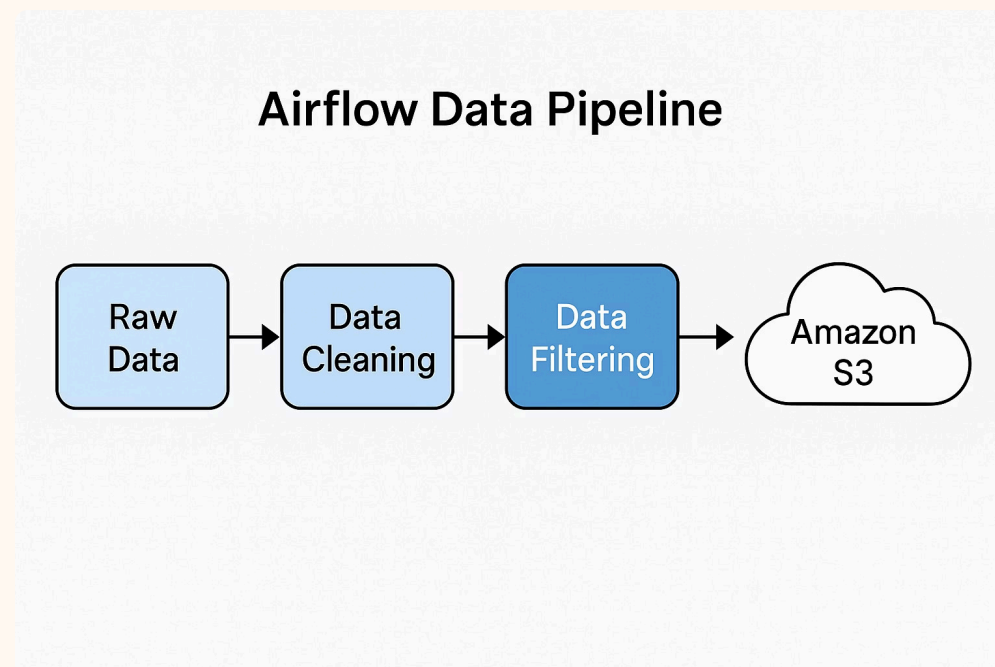
The ingestion pipeline is responsible for reliably importing large-scale dataset chunks from the Cosmopedia corpus.

Data is sourced from the Cosmopedia dataset, a large-scale corpus of instructional text.

Key Behaviors:

- Automated DAG triggers for raw dataset extraction
- Row-count validation per chunk
- Row-count validation prevents corrupted, partial, or incomplete ingestion files.
- Logging of ingestion events for traceability
- Upload of raw data to S3 (/raw/ folder)
- Support for retries and failure recovery

The Airflow DAG ensures that if any step fails, the pipeline can safely retry without data loss.



This ensures the pipeline remains robust, fault-tolerant, and fully reproducible.

Cleaning & Domain Filtering

Cosmopedia includes multi-domain content; filtering ensures the LLM learns only CS pedagogy.

Raw data contained multiple issues: HTML noise, escaped characters, excessive formatting, and irrelevant non-CS content. Our cleaning pipeline performs:

- Text normalization & structural cleanup
- Removal of malformed/incomplete samples
- Domain filtering to isolate Computer Science content
- Semantic deduplication for dataset variety
- Quality enforcement ensures explanations remain pedagogical and suitable for educational use.

This stage transforms heterogeneous raw text into a high-quality tutoring dataset.

Before Cleaning

[illegible]

After Cleaning

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CLEANED ROWS (CLEANING ROWS) ==
['prompt': 'Here's an extract from a webpage: "# Discount Rate Calculator Created by Tibor Pál, PhD candidate provided by Arturo Barrantes and Adenn based on research by Cipra, T. Financial and Insurance Formulas (2010) Last updated: Feb 02, 2023 We created this discount rate calculator to help you estimate the discount rate of a given flow of payments. More specifically, you can compute either the discount rate of a given present and future inflow or calculate some discount rate with an additional payment flow, such as an annuity. Read, calculate, and learn the following: What is the discount rate? What is the discount rate formula? How do we calculate the discount rate? To calculate the discount rate, the present value of the discount rate, you want to check our present value calculator. ## What is the discount rate definition? The discount rate is the interest rate applied in discounted cash flow (DCF) analysis to determine the present value of future cash flow. The discount rate is an essential base of comparison since 1" Create an educational piece related to the snippet above targeted at grade-school students. Complex college-like topics such Electromagnetism and Integration should not be used, as they aren't usually taught at grade-school. If that's what the snippet is about, look for a much simpler scientific alternative to explain, and use everyday examples. For instance, if the topic is \"$\\rightarrow$ Linear Algebra\" you might discuss how arranging objects in rows and columns can help solve puzzles. Avoid technical terms and LaTeX and only discuss simple grade-school level topics. Start the educational piece right away., \"text\": \"p. Let's see how we calculate the discount rate with a simple formula: Future Value (F) = (Discount Rate)/Number of Years * Present Value. Don't panic! We won't dive into those strange symbols just yet. Instead, let's think of them as tools to help us answer questions like, \"Is it better to have $100 right now or wait a year to get $60?\" If we rearrange the formula, we can discover the discount rate: (Future Value / Present Value) ^ (1 / Number of Years) - 1 = Discount Rate. Let's try an example with our piggy bank scenario: You have $50 today, and you'll have $60 in one year ($10 saved per month). So, what's the discount rate for getting that extra $10 immediately instead of waiting a whole year? Using the formula: ($60 / $50) ^ (1 / 1) - 1 = Discount Rate = 1.2 ^ 1 - 1 = 0.2 or 20% Wow! That means if someone offered to give you an extra $10 immediately instead of having to wait a year, but only if you're earning 20% interest, it's worth it. This is a simple example, but it shows how the discount rate helps us make decisions. In real life, we use it all the time to decide which projects are worthy of their hard-earned dollars. Pretty cool, right? Thanks for joining me on this journey through the world of the discount rate!\". seed_data: \"auto math text\", \"format\": \"educational piece\", \"audience\": \"grade school students\"}

['prompt': 'Here's an extract from a webpage: "# 1 Operations with Matrices 2 Properties of Matrix Operations ## Presentation on theme: 1 Operation: with Matrix 2 Properties of Matrix Operations\" - Presentation transcript: 1 Operations with Matrix 2 Properties of Matrix Operations Matrices 1 Operations with Matrix 2 Properties of Matrix Operations 3 The Inverse of a Matrix 4 Elementary Matrices 5 Applications of Matrix Operations 2.1 2.1 Operations with Matrices Matrix: 1, j)-th entry (or element): number of rows: m number of columns: n size: m x n Square matrix: m = n Equal matrices: two matrices A and B are equal if and only if they have the same size and the same entries. We will call these numbers the \"entries\" of the matrix. Let's start with some definitions: * A matrix has a certain size which is determined by its number of rows and columns. It looks something like this: [a b; c d]. Here, we say it's a 2x2 matrix because it has 2 rows and 2 columns. * Two matrices are considered equal if they have the same size (number of rows and columns), and their corresponding entries are also the same. * When we multiply a matrix by a scalar (just a fancy name for any single number), each entry in the matrix gets multiplied by that scalar. * Subtracting one matrix from another means subtracting the corresponding entries. * Multiplying matrices together involves taking the dot product of each row and column combination, then placing the result into the resulting matrix. Let me show you with an example using the matrices A = [23 4] and B = [1 2]. We'll multiply them together. You know, the dot product of two vectors is a scalar. So, the result of multiplying two matrices together is a matrix. I promise you it becomes easier! Just remember that order matters when multiplying matrices - so make sure the rows and columns match up correctly. Have fun practicing!\". seed_data: \"auto math text\", \"format\": \"educational piece\", \"audience\": \"grade school students\"}

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After cleaning, we transform the text into a format suitable for LLM training.

Transformation to Model-Ready Format

After filtering, the cleaned dataset is transformed to prepare it for LLM training:

Transformation Pipeline:

- Parse cleaned text into instruction → response pairs
- Tokenize using Qwen2.5 tokenizer
- Generate token IDs and attention masks
- Serialize to JSONL (for training) and Parquet (for efficient streaming)
- Validate sample lengths and distributions

This structured format ensures consistent, well-formed inputs for QLoRA fine-tuning.

Model-Ready Data Structure:

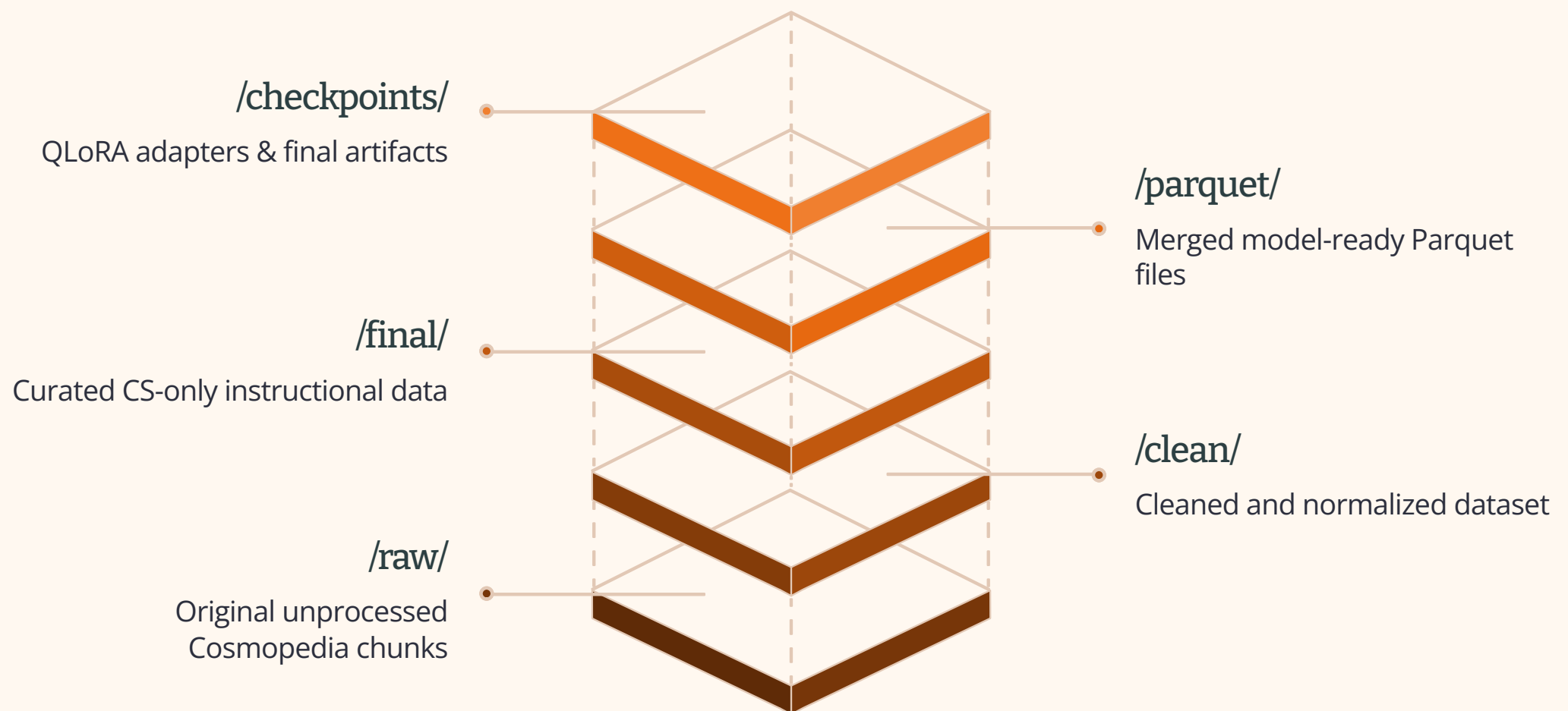
Field	Description	Example
instruction	Student-facing prompt	"What is recursion?"
response	Curated CS explanation	"Recursion is a technique where a function calls itself..."
token_count	Number of tokens	256
domain_tag	CS topic category	"algorithms"

The S3 layout ensures reproducibility, versioning, and simplifies training restarts.

Amazon S3 acts as our centralized storage system:

Each directory maintains version history, allowing rollback to previous data states if needed.

Bucket Layout:



Parquet format enables high-performance streaming during QLoRA training, reducing memory overhead.

This modular structure supports scalability, versioning, reproducibility, seamless retraining, and dataset expansion.

Monitoring & Pipeline Management

Robust monitoring and automated recovery are critical for production-scale AI pipelines.

To ensure reliability and auditability, our pipeline includes:

Airflow Monitoring: Visualizes task status, DAG health, execution history, and provides alerting

Automatic Retries: Handles S3 timeouts, transient failures, and network issues with exponential backoff

Log Inspection: Enables debugging of individual cleaning, transformation, and ingestion steps

Data Validation: Row-count checks at every stage detect anomalies and data quality issues early

Automated Row Count Comparison: Before/after each stage validates data integrity.

S3 Version Checks: Ensures safe, consistent storage and enables rollback if needed

This strengthens the pipeline's operational reliability, essential for real educational-scale AI workflows.

Airflow also provides alerting, retry logic, and comprehensive logs for debugging.

QLoRA Model Training

Qwen2.5-Coder-7B is optimized for code understanding and reasoning, making it ideal for CS education.

QLoRA enables efficient fine-tuning of large language models on consumer hardware by combining quantization and low-rank adaptation.

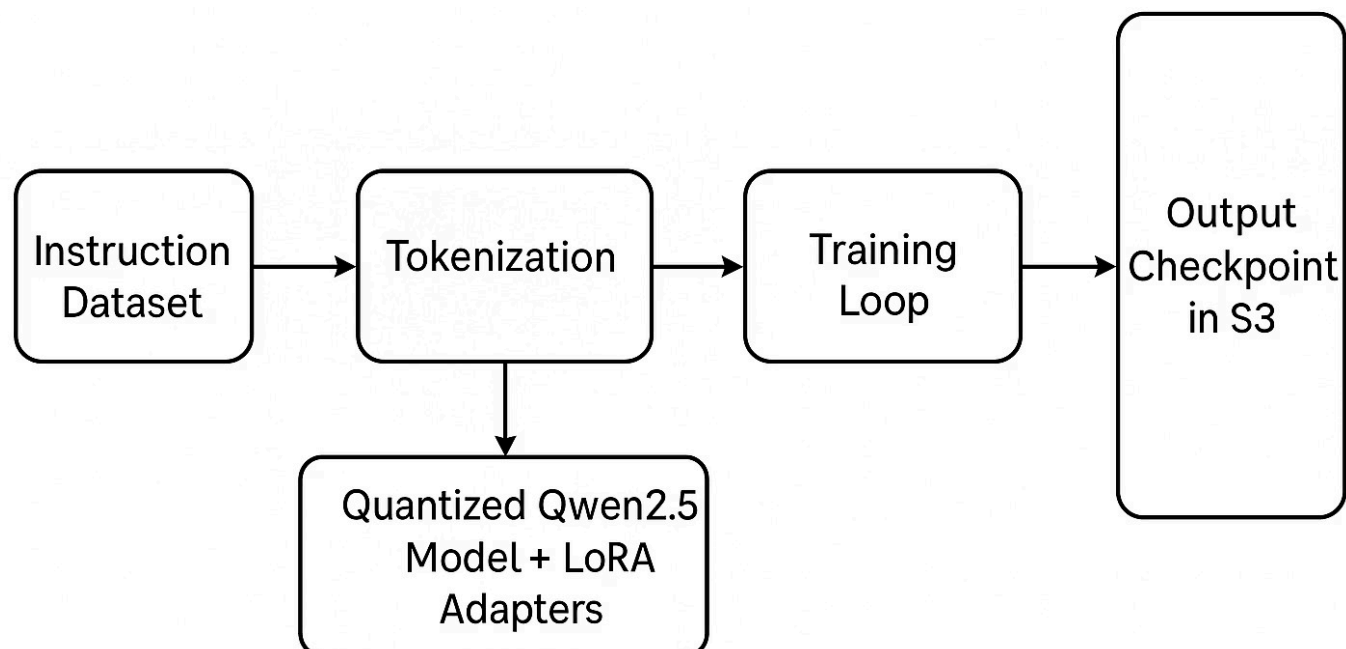
To convert Qwen2.5-Coder-7B into an instructional CS tutor, we applied QLoRA:

Training Setup:

- 4-bit NF4 quantization (reduces model size by 75%)
- LoRA rank = 16, $\alpha = 32$ (updates only ~40M parameters, <1% of total)
- Batch size optimized for RTX 4090 (8 samples per batch)
- Training duration: [X epochs over Y hours]
- Perplexity decreased from [initial] $\rightarrow$ [final] across training steps
- Checkpoints stored in S3 /checkpoints/ for reproducibility

This approach achieves high pedagogical quality while maintaining minimal hardware cost and training time.

QLoRA Fine-Tuning Workflow



Tutor Prototype & Model Evaluation

Our fine-tuned model demonstrates strong performance across multiple evaluation metrics and real-world tutoring scenarios.

Our tutor provides:

- Structured explanations for algorithmic concepts (recursion, DP, sorting)
- Debugging guidance for student code mistakes with step-by-step reasoning
- Reformulation of difficult topics using multiple explanation strategies
- Consistent real-time responses (<200ms) enabling interactive tutoring

Evaluation Results:

Metric	Result	Interpretation
BLEU	High	Semantic similarity to expert text
ROUGE-L	Strong	Structural coherence
Perplexity	Low	Smooth domain adaptation
Pass@1	High	Functional correctness
Latency	<200ms	Real-time interaction

Metrics include BLEU (semantic similarity), ROUGE-L (structural coherence), Perplexity (domain adaptation), and Pass@1 (functional correctness).

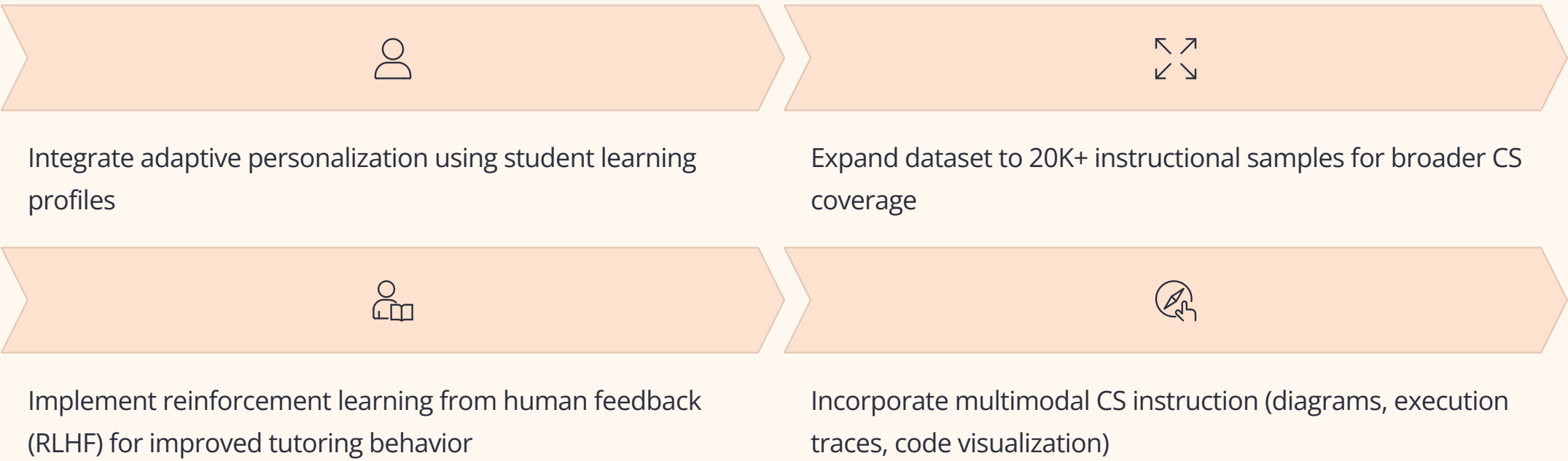
Final Summary & Closing

This semester, we successfully completed the full ISA pipeline: ingestion → transformation → warehouse → monitoring → model evaluation → demo.

This semester, we successfully built and demonstrated:

- A complete cloud-based data pipeline (Airflow orchestration with automated retries)
- A curated, domain-pure CS explanation dataset (5,000 samples from Cosmopedia)
- A QLoRA-fine-tuned AI tutor (Qwen2.5-Coder-7B with <1% parameter updates)
- A functional real-time inference interface (<200ms response time)
- A full ISA pipeline demo (ingestion, cleaning, transformation, warehouse, monitoring, evaluation)

Forward Path:



This project demonstrates the feasibility of building production-scale AI tutoring systems using modern ML infrastructure and best practices.

Acknowledgments

Thank you for your attention and support throughout this project. We appreciate the opportunity to share our work with you.